

FUNCTIONAL REQUIREMENT DOCUMENT (FRD)

Recommended For You – Inventory Liquidation Engine

Organization: Snitch

Document Type: Functional Requirement Document

Prepared By: Business Analyst

Version: 1.0

Date: April 2026

Author, Change Control

Date	Author	Version	Description
2-04-2026	BA	1.0	Initial FRD

Approvers:-

Approver	Department	Signature and Date
Project Sponsor	Management	
Product Head	E-commerce	
Tech Head	Technology	

Table of Contents:-

1. Introduction	2
1.1 Purpose	2
1.2 Scope	2
2. System Overview	2
Key Capabilities	2
3. Stakeholders	2
4. Functional Requirements	3
4.1 Recommendation Engine Module	3
4.1.1 Product Selection Engine	3
4.1.2 Ranking & Sorting Engine	4
4.1.3 Product Display Module	4
4.1.4 Navigation & UI Module	5
4.1.5 Admin Control Module	5
4.1.6 Offer Integration Module	5
4.1.7 Data Refresh & Scheduler	5
5. Non-Functional Requirements	6
Security	6
Performance	6
Scalability	6

6. Assumptions	6
7. Future Enhancements	6

1. Introduction

1.1 Purpose

This FRD defines the detailed functional and system-level requirements for the **“Recommended For You” Inventory Liquidation Engine** on the Snitch website. The document outlines system behavior, workflows, modules, and logic required to enable inventory-driven product recommendations.

1.2 Scope

- The system will support:
- Inventory-driven recommendation logic
 - Product ranking based on business rules
 - Discount-focused product display
 - Admin configuration and control
 - UI integration (homepage + sidebar)
 - Offer and pricing integration
 - Scheduled or real-time updates

2. System Overview

The system acts as a **recommendation layer** within the e-commerce platform focused on clearing slow-moving inventory.

Key Capabilities

- Automated product selection based on inventory data
- Rule-based ranking and prioritization
- Dynamic UI rendering of recommended products
- Admin override and configuration controls
- Integration with pricing and discount systems
- Scheduled refresh of product lists

3. Stakeholders

Stakeholder	Role
Product Team	Feature definition
Inventory Team	Inventory data management

Stakeholder	Role
Marketing Team	Discount strategy
Tech Team	Development & maintenance
Admin Team	Product tagging and configuration
Customers	End users

4. Functional Requirements

4.1 Recommendation Engine Module

4.1.1 Product Selection Engine

Description

This module identifies eligible products for recommendation based on inventory and sales data.

Functional Requirements

The system shall:

- Fetch product data from inventory database
- Identify products where:
 - Inventory age > configured threshold
 - Stock quantity is above defined level
 - Sales velocity is below defined threshold
- Include manually tagged products regardless of criteria
- Exclude out-of-stock products
- Ensure only active products are displayed

Data Fields

Field	Description
Product ID	Unique identifier
Inventory Age	Days since added
Stock Quantity	Available stock
Sales Velocity	Units sold per time
Discount Value	Applied discount
Admin Tag	Manual tag indicator

4.1.2 Ranking & Sorting Engine

Description

Determines the order in which products are displayed.

Functional Requirements

The system shall:

- Prioritize products based on:
 1. Admin priority tag
 2. Inventory age (descending)
 3. Stock quantity (descending)
 4. Discount percentage (descending)
- Allow configurable weightage (future scope)
- Ensure no duplicate product entries

4.1.3 Product Display Module

Description

Handles UI rendering of recommended products.

Functional Requirements

The system shall:

- Display product cards with:
 - Product image
 - Product name
 - Discount badge (e.g., “40% OFF”)
 - Tags (Clearance, Hot Deal, Limited Stock)
 - Original price
 - Discounted price
- Highlight high-priority products visually
- Limit number of displayed products (configurable)

4.1.4 Navigation & UI Module

Description

Manages placement and accessibility of the recommendation section.

Functional Requirements

The system shall:

- Display “Recommended For You” section on homepage
- Place section above fold or near top
- Introduce left sidebar navigation
- Move existing navigation to top bar
- Provide clickable access via sidebar tab

4.1.5 Admin Control Module

Description

Provides administrative control over product selection and display.

Functional Requirements

The system shall allow admin users to:

- Tag products as:
 - Recommended
 - Clearance
 - High Priority
- Configure:
 - Inventory age threshold
 - Stock threshold
 - Sales velocity threshold
 - Display count limit
- Override automated selection logic
- Enable/disable products from recommendation

4.1.6 Offer Integration Module

Description

Integrates promotional offers into recommended products.

Functional Requirements

The system shall:

- Apply coupons to eligible products
- Support flash deals with time limits
- Enable bundle pricing for grouped products
- Display offer indicators on product cards

4.1.7 Data Refresh & Scheduler

Description

Ensures product list remains updated.

Functional Requirements

The system shall:

- Refresh product list:
 - In real-time OR
 - At scheduled intervals (e.g., every 6 hours)
- Recalculate ranking on refresh
- Remove out-of-stock products automatically
- Log refresh activity

5. Non-Functional Requirements

Security

- Admin access must be role-based
- Data integrity must be maintained
- Secure API communication

Performance

- Page load impact must be $< 10\%$
- Recommendation section load time < 2 seconds

Availability

- System uptime $\geq 99\%$
- Fail-safe fallback (hide section if error occurs)

Scalability

- Must support large SKU catalog
- Should handle high traffic without performance degradation

6. Assumptions

- Inventory and sales data are accurate
- Admin users will actively manage tagging
- Discount system is already implemented
- Users have access to updated website UI

7. Future Enhancements

- AI/ML-based personalized recommendations
- User behavior tracking integration
- A/B testing for recommendation strategies
- Predictive inventory liquidation models
- Integration with marketing channels